

Why dividing by n gives a biased estimator of variance

Suppose $X_1, \dots, X_n$ are i.i.d. with

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Consider the estimator

$$\hat{s}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

We want to show that $\hat{s}^2$ is a biased estimator of the population variance σ^2 .

An estimator is unbiased if

$$E[\hat{s}^2] = \sigma^2.$$

So our goal is to compute $E[\hat{s}^2]$.

Step 1: Algebra – simplify the sum of squared deviations

Start with

$$\sum_{i=1}^n (X_i - \bar{X})^2.$$

Expand the square:

$$\sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2).$$

Distribute the summation:

$$= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2.$$

Since

$$\sum_{i=1}^n X_i = n\bar{X}$$

and $\bar{X}^2$ does not depend on i ,

$$\sum_{i=1}^n \bar{X}^2 = n\bar{X}^2.$$

Therefore,

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n X_i^2 - 2\bar{X}(n\bar{X}) + n\bar{X}^2 \\ &= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2. \end{aligned}$$

Hence,

$$\boxed{\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2}$$

and therefore

$$\hat{s}^2 = \frac{1}{n} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right).$$

Step 2: Find $E[X_i^2]$

Recall the variance identity:

$$\text{Var}(X_i) = E[X_i^2] - E[X_i]^2.$$

Since

$$\text{Var}(X_i) = \sigma^2$$

and

$$E[X_i] = \mu,$$

we have

$$\sigma^2 = E[X_i^2] - \mu^2.$$

Therefore,

$$\boxed{E[X_i^2] = \sigma^2 + \mu^2}.$$

Step 3: Expand $\bar{X}^2$

By definition,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

So

$$\bar{X}^2 = \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2.$$

Pull out $1/n^2$:

$$\bar{X}^2 = \frac{1}{n^2} \left(\sum_{i=1}^n X_i \right)^2.$$

When we square a sum, we get the squared terms plus the cross-product terms:

$$\boxed{\bar{X}^2 = \frac{1}{n^2} \left(\sum_{i=1}^n X_i^2 + 2 \sum_{i < j} X_i X_j \right)}.$$

Step 4: Take expectations of $\bar{X}^2$

Taking expectations,

$$E[\bar{X}^2] = \frac{1}{n^2} \left[\sum_{i=1}^n E[X_i^2] + 2 \sum_{i < j} E[X_i X_j] \right].$$

For the squared terms,

$$E[X_i^2] = \sigma^2 + \mu^2.$$

There are n such terms, so

$$\sum_{i=1}^n E[X_i^2] = n(\sigma^2 + \mu^2).$$

For the cross-product terms, if $i \neq j$, independence implies

$$E[X_i X_j] = E[X_i]E[X_j].$$

Since each observation has mean μ ,

$$E[X_i X_j] = \mu^2.$$

How many cross-product terms are there?

$$\sum_{i < j}$$

contains

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

pairs. Because there is a 2 in front,

$$2 \cdot \frac{n(n-1)}{2} = n(n-1).$$

Therefore,

$$E[\bar{X}^2] = \frac{1}{n^2} \left[n(\sigma^2 + \mu^2) + n(n-1)\mu^2 \right].$$

Expand:

$$= \frac{1}{n^2} \left[n\sigma^2 + n\mu^2 + n^2\mu^2 - n\mu^2 \right].$$

So

$$= \frac{1}{n^2} \left[n\sigma^2 + n^2\mu^2 \right].$$

Hence,

$$\boxed{E[\bar{X}^2] = \mu^2 + \frac{\sigma^2}{n}}.$$

Step 5: Put everything together

Recall that

$$\hat{s}^2 = \frac{1}{n} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right).$$

Take expectations:

$$E[\hat{s}^2] = \frac{1}{n} \left[\sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2] \right].$$

Substitute the results from Steps 2 and 4:

$$E[\hat{s}^2] = \frac{1}{n} \left[n(\sigma^2 + \mu^2) - n \left(\mu^2 + \frac{\sigma^2}{n} \right) \right].$$

Expand:

$$= \frac{1}{n} \left[n\sigma^2 + n\mu^2 - n\mu^2 - \sigma^2 \right].$$

The μ^2 terms cancel:

$$= \frac{1}{n} \left[n\sigma^2 - \sigma^2 \right].$$

Factor out σ^2 :

$$= \frac{1}{n}(n-1)\sigma^2.$$

Therefore,

$$E[\hat{s}^2] = \frac{n-1}{n}\sigma^2.$$

Since

$$\frac{n-1}{n} < 1,$$

we have

$$E[\hat{s}^2] < \sigma^2.$$

Thus,

$$\hat{s}^2 \text{ is downward biased for } \sigma^2.$$

Its bias is

$$\begin{aligned} \text{Bias}(\hat{s}^2) &= E[\hat{s}^2] - \sigma^2 \\ &= \frac{n-1}{n}\sigma^2 - \sigma^2 \\ &= \boxed{-\frac{\sigma^2}{n}}. \end{aligned}$$

Why do we divide by $n-1$ instead?

Because

$$E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n-1)\sigma^2.$$

So if we define

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

then

$$E[s^2] = \sigma^2.$$

That is why the usual sample variance uses $n-1$ rather than n .